Cailyn Smith

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RESEARCH STATEMENT

My research interests broadly lie in how robots can learn more effectively from less human data, especially for unstructured manipulation tasks. I am interested in accounting for suboptimal human demonstrators, learning from multiple demonstrators, and learning with multimodal inputs.

EDUCATION

Carnegie Mellon University, Pittsburgh, PA Jan 2026 - Present
Ph.D. Student in Robotics, NSF Graduate Research Fellow *GPA: 4.0 / 4.0*
Advisors: Dr. Henny Admoni and Dr. Zackory Erickson

Colorado School of Mines, Golden, CO August 2021 - May 2025
B.S. Computer Science, focus in Robotics & Intelligent Systems. Minor: Physics. *GPA: 3.99 / 4.0*

Relevant Coursework: Robot Learning; Mobile Manipulation; Robot Kinematics, Dynamics, & Control; Numerical Optimization; Robot Mapping & Localization; Computer Vision; Advanced ML

RESEARCH EXPERIENCE

RCHI Lab and HARPLab @ Carnegie Mellon Jan 2026 - Present
PhD Student

- Developed an interactive imitation learning algorithm that fine-tunes diffusion policies via blended shared control, improving autonomous performance > 75% over HG-DAgger on real-world long-horizon manipulation
- Leading a team of five students to build an autonomous manipulation stack for robot-assisted feeding, spanning articulated-object interaction, button pressing, container opening, and pick-and-place

Interactive Robotics Research Lab @ Colorado School of Mines Jun 2020 - May 2025
Undergraduate Researcher

- Developed reinforcement learning algorithms to dynamically change a robot's autonomy during interactions
- Designed and conducted human-subjects experiments on level of autonomy and social norms in robotics

Autonomy, Robotics, & Intelligent Algo. Lab @ Colorado School of Mines Dec 2023 - May 2025
Undergraduate Researcher

- Evaluated state-of-the-art SLAM algorithms on a novel benchmark dataset to characterize their robustness
- Contributed towards a journal paper in preparation for submission to IEEE Transactions on Robotics (T-RO)

Autonomous Robotics and Control Lab @ Caltech Jun 2024 - Aug 2024
WAVE Fellow

- Fine-tuned spacecraft segmentation neural network models to utilize long-wave thermal space imagery
- Deployed onboard training algorithms on satellite flight hardware using Docker and ROS2 (launched in 2025)

PUBLICATIONS

1. **Cailyn Smith**, Henny Admoni, and Zackory Erickson. "BlenDAgger: Blended Shared Control for Interactive Imitation Learning." In submission.
2. **Cailyn Smith**[1], Lara Bezerra[1], Rafael Sousa Silva, Mark Higger, and Tom Williams. "Evaluating an Adaptive Model of Robot Performative Autonomy for Human-Robot Collaboration." In submission.
3. Terran Mott[1], **Cailyn Smith**[1], Aaron Fanganello, Tom Williams. "We Don't Ask Such Things: A Mixed-Methods Reassessment of Bounded Proportionality in Robotic Norm Violation Response." In: International Journal of Social Robotics. 2026.
4. Rafael Sousa Silva, **Cailyn Smith**, Lara Bezerra, Tom Williams. "Evaluating Robotic Performative Autonomy In Latency-Sensitive Collaborative Contexts." In: IEEE Int'l Conf. on Robotics and Automation (ICRA). 2025.
5. **Cailyn Smith**, Terran Mott, and Tom Williams. "Robot, Take the Joystick: Understanding Space Robotics Experts' Views on Autonomy." In: IEEE Int'l Symp. on Human-Robot Interactive Communication (RO-MAN). 2024.

6. **Cailyn Smith**, Terran Mott, and Tom Williams. "Perspectives on Level of Autonomy Decisions in Space Robotics." In: Companion Proc. of the ACM/IEEE Int'l Conf. on Human-Robot Interaction (HRI): Late Breaking Reports. 2024.
7. Rafael Sousa Silva, **Cailyn Smith**, Lara Bezerra, Tom Williams. "Toward RAPS: the Robot Autonomy Perception Scale." In: Variable Autonomy for Human-Robot Teaming Workshop at the IEEE Int'l Symp. on Human-Robot Interactive Communication (RO-MAN). 2024.
8. **Cailyn Smith**[1], Charlotte Gorgemans[1], Ruchen Wen, Saad Elbeleidy, Sayanti Roy, and Tom Williams. "Leveraging Intentional Factors and Task Context to Predict Linguistic Norm Adherence." In: Annual Meeting of the Cognitive Science Society (CogSci). 2022.

[1] Authors contributed equally to this work

INDUSTRY EXPERIENCE

Autonomous Solutions Inc. Aug 2025 - Dec 2025
Software Engineer

· Developed multi-vehicle command & control software for autonomous construction and agriculture vehicles

United Launch Alliance May 2023 - Aug 2023
Software Engineering Intern

· Developed software in Python and C++ to perform closed-loop Monte Carlo rocket trajectory simulation

Katalyst Space Technologies May 2022 - Aug 2022
Software Engineering Intern

· Designed perception algorithm architectures and behaviors for tracking Resident Space Objects

SCHOLARSHIPS & RECOGNITIONS

NSF Graduate Research Fellow (2025): National fellowship for research in STEM fields
CRA Outstanding Undergraduate Researcher Runner-Up (2025): Top 16 nationally
Outstanding Graduating Senior (2025): Awarded to top 2 computer science undergraduates at Mines
Colorado Engineering Council Silver Medal Award (2025): Awarded to one graduating student at Mines
Astronaut Scholar (2023 & '24): Nationally competitive scholarship awarded to ∼70 students nationwide
Florence Caldwell Full Tuition Scholar (2021-25): Awarded to 3 students annually at Mines

SELECTED PROJECTS

Disambiguating Manipulation Requests Jan 2026 - May 2026
Mobile Manipulation (ROB 762)

· Implemented a bidirectional communication system using VLMs to request more information until the user's request is unambiguous. Evaluated on mobile manipulation pick and place tasks with a Stretch 3 robot

Reinforcement Learning for Manipulation Jan 2026 - May 2026
Robot Learning (ROB 831)

· Implemented on-policy, off-policy, and model-based RL algorithms for manipulation in RoboSuite

Autonomous Rover Aug 2021 - Dec 2023
NASA's Colorado Space Grant Robotics Challenge Robotics Team

· Implemented object detection with ROS2 and Python using image segmentation on depth images

RL for Robot Navigation Jan 2023 - May 2023
Robotic Perception & Programming (CSCI 573)

· Coded on-policy and off-policy reinforcement learning algorithms for LiDAR-based navigation with ROS

SKILLS

Coding: Proficient in Python, C++, C#, Java, MATLAB. Knowledge of C, R, Assembly, Bash.
Software: TensorFlow, PyTorch, ROS1 & ROS2, MuJoCo, RoboCasa, RoboSuite, Docker
Robots: Kinova Gen3, UFACTORY xArm, Franka Panda, Hello Robot Stretch, Misty II, SoftBank Nao